Comprehensive Intelligence Architecture and Data Infrastructure for the Jordanian Real Estate Market

The development of an exclusive, institutional-grade real estate intelligence platform for the Hashemite Kingdom of Jordan requires a rigorous mapping of the country's highly fragmented, yet rapidly digitizing, data ecosystem. The Jordanian real estate market is characterized by a unique structural dichotomy: while official government infrastructure provides robust macroeconomic signals and highly granular spatial (GIS) frameworks, it maintains strict statutory confidentiality regarding unit-level transaction prices. Consequently, achieving true market intelligence—and ultimately deploying Automated Valuation Models (AVMs)—necessitates the synthesis of official cadastral records, municipal zoning APIs, macroeconomic indicators, and scraped portal data to reverse-engineer market realities.
This report systematically deconstructs every identifiable data vector within Jordan, providing the exact endpoints, extraction protocols, and analytical context required to ingest this data into a centralized intelligence platform. The analysis indicates that Jordan is aggressively modernizing its digital infrastructure, evidenced by extensive deployments of ESRI ArcGIS REST services across municipal and federal bodies, and the adoption of CKAN frameworks for open data publishing. By systematically harvesting these endpoints, an intelligence platform can construct a foundational database capable of supporting hedonic regression, repeat sales indices, and residual land value calculations.

The Cadastral and Legal Data Foundation

The bedrock of any real estate intelligence platform is the authoritative ledger of land ownership, boundaries, and legal encumbrances. In Jordan, this domain is governed primarily by the Department of Lands and Survey (DLS) and the Ministry of Justice (MoJ).

Department of Lands and Survey (DLS)

The DLS is the definitive source for property registration in Jordan. While individual transaction prices are shielded by privacy laws, the DLS provides vital baseline data including cadastral boundaries, expropriation announcements, and aggregated transaction volumes. The recent launch of the DLS e-services portal represents a significant digitization effort, allowing registered users to query specific plots via coordinates, retrieve ownership statuses, and access administrative property valuation records.
Furthermore, the DLS publishes data to the national Open Data Portal (powered by CKAN), exposing a robust Data API (datastore_search) that allows direct JSONP/Python queries against state datasets. These datasets include expropriation announcements, tenders, and directories of licensed surveyors. For an intelligence platform, the administrative valuations provided by the DLS serve as a baseline for tax and fee calculations, though they must not be mistaken for achieved market prices.
To extract deeper intelligence, the platform operator may leverage Jordan's Access to

Information Law No. 47 (amended 2024), which provides a legal mechanism to formally request anonymized, aggregate transaction data at the neighborhood level directly from the DLS Director General's office.

Ministry of Justice (MoJ) and Ministry of Finance (MoF)

A critical secondary source for pricing floors is the Ministry of Justice's electronic auction portal. Launched to modernize the execution of court-ordered sales, this portal lists distressed properties and foreclosures, providing a rare window into baseline asset valuations, appraisal reports, and forced-liquidation pricing. The transition to electronic bidding has reportedly increased bidder participation by four to five times, making this a highly liquid subset of the market.

Concurrently, the Ministry of Finance (MoF), in collaboration with the Ministry of Digital Economy and Entrepreneurship, operates a vocational licensing database developed under the Property Tax Project. This system tracks commercial entities and their locations, providing an essential layer of commercial real estate utilization and retail footprint data.

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
Department of Lands and Survey (DLS) Main Portal	https://www.dls.gov.jo/Default/En	National	Monthly real estate trading volumes, transaction counts, annual statistical yearbooks, cadastral maps	Governorate / Plot	Manual PDF parsing / OCR	No	None	Monthly / Annual	None	Standard web limits	Provides volume/activity indices. Reports published in Arabic require OCR/translation pipelines.
DLS E-Services	https://eservices.dls.gov.jo	National	Ownership status, encumbrance inquiry, property valuation	Plot level	Web Scraping / Selenium	No	Required (National ID/Registration)	Live	None	Session timeouts, Captchas likely	Essential for due diligence on specific parcels. Administrative

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
			records, plot coordinates								valuations are for tax purposes.
Jordan Open Data Portal (CKAN) - DLS Datasets	https://opendata.gov.jo/en/api/3/action/datastore_search?resource_id=1fd227b0-deba-493f-a88d-1e8f3ffcc6c9	National	Expropriation announcements, real estate newsletters, DlsVillageCode, licensed surveyors	Governorate	CKAN API / JSON	Yes	None	Periodic	https://opendata.gov.jo/en/api-guide/	Standard CKAN API rate limits	Accesses DLS datasets hosted on the national CKAN registry. Data returned in JSON format.
Ministry of Justice (MoJ) E-Auctions	http://moj.gov.jo	National	Foreclosures, public auctions, property specifications, appraisal reports, auction dates	Unit / Plot level	Web Scraping / Playwright	No	Required for bidding; public for viewing	Daily	None	IP rate limiting	Crucial for identifying distressed asset pricing. Requires scraping dynamic web views.
Ministry of Finance (MoF)	https://mof.gov.jo/Default/EN/	National	Public revenues, property tax policies, vocatio	Governorate / Sector	Manual request / Web Scraping	No	None	Annual / Ad-hoc	None	Standard web limits	Relevant for understanding broader taxation framew

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
			nal licensing								orks affecting property yields.

Municipal Spatial Data Infrastructures (SDI) and Zoning

The regulatory layer of real estate—specifically what can legally be built on a plot—is the primary driver of residual land value. A platform that relies solely on transaction data without understanding the underlying zoning constraints is fundamentally flawed. In Jordan, spatial data infrastructure is relatively advanced, with multiple authorities publishing ArcGIS REST services.

Greater Amman Municipality (GAM)

The GAM operates the most sophisticated Geographic Information System (GIS) in the country, exposing public ArcGIS REST services that detail zoning classifications across the capital. These classifications (Residential A, B, C, D, Commercial, Industrial, Agricultural) define permitted floor area ratios (FAR), maximum heights, setbacks, and permitted uses. This is the single most valuable spatial dataset for real estate analysis in Jordan. By leveraging spatial joins via the ArcGIS API, an intelligence platform can automatically intersect scraped property listings with GAM zoning polygons, instantly assigning regulatory constraints to raw market data. The GAM endpoints (ammancitygis.gov.jo) also expose layers for street names, building numbers, traffic safety systems, and infrastructure networks. A deeper analysis of these layers enables the identification of urban mobility hotspots, which are highly correlated with commercial retail premiums.

ASEZA, PDTRA, and MoLA

Outside Amman, spatial data is governed by specialized regional authorities and the Ministry of Local Administration (MoLA). The Aqaba Special Economic Zone Authority (ASEZA) maintains an independent geoportal detailing land use master plans, tourism zoning, and logistics corridors for Jordan's sole coastal city. Similarly, the Petra Development and Tourism Region Authority (PDTRA) exposes ArcGIS REST endpoints mapping protected valleys, administrative offices, and development zones critical for hospitality investments.
For the remainder of the country, MoLA governs municipal boundaries and applies the *Al-Taqseem* subdivision laws. Under this framework, one-third of subdivided land is designated for public purposes, which heavily impacts net developable area. MoLA's e-services portal, integrated with the national Sanad application, processes property clearance certificates—a leading indicator of transaction intent.

The Role of OpenStreetMap

Where official GIS endpoints are restricted or incomplete, OpenStreetMap (OSM) serves as a critical fallback. The Jordan extract, updated daily via Geofabrik, provides physical geography, road networks, building footprints, and points of interest (POIs) across all Jordanian cities. While OSM lacks regulatory zoning classifications (showing only what physically exists rather than what is legally permitted), it is indispensable for calculating proximity metrics (e.g., distance to schools, hospitals, or transit nodes) which are required variables in any hedonic pricing model.

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
Greater Amman Municipality (GAM) ArcGIS REST	https://www.ammancitygis.gov.jo/arcgis/rest/services	Amman	Zoning classifications (Residential A-D, Commercial), land use, road networks, building footprints, POIs	Plot / Parcel level	ArcGIS REST / Python arcgis	Yes	None for public layers	Continuous	https://www.ammancitygis.gov.jo/arcgis/rest/info	Connection throttling on heavy spatial queries	The single most valuable spatial dataset. Allows programmatic intersection of scraped listings with zoning polygons.
Ministry of Public Works and Housing (MPWH) GIS	https://gis.mpwh.gov.jo/arcgis/rest/services	National	Infrastructure projects, national road networks, public buildings, topographic layers	Street / Plot level	ArcGIS REST / WFS	Yes	None	Periodic	https://gis.mpwh.gov.jo/arcgis/rest/info	Standard ESRI server limits	Vital for modeling accessibility premiums and transport infrastructure impact on land values.
PDTRA Spatial	https://sp.pdtr	Petra Region	Regional	Plot / Zone	ArcGIS REST	Yes	None for	Ad-hoc	https://sp.pdtr	Unknown	Critical for

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
Services	a.gov.jo/arcgis/rest/services		zoning, valley boundaries, administrative offices, protected areas	level			base layers		a.gov.jo/arcgis/rest/info		tourism-related real estate investment modeling in southern Jordan.
ASEZA Geoportal (Aqaba)	ht[span_34](start_span)[span_34](end_span)tps://www.arcgis.com/home/item.html?id=d7f591c9dff647068e01de91988048fa	Aqaba Zone	General land use, Iqlim zoning tables, district administrative boundaries	District / Zone level	ArcGIS REST / JSON	Yes	None	Periodic	None	ArcGIS Online standard limits	Details the master plan for Aqaba's special economic zone, including tourism and logistics zoning.
Ministry of Local Administration (MoLA) E-Services	https://eservices.mola.gov.jo	National (excl. Amman/Aqaba)	Municipal boundaries, subdivision plans (Al-Taqseem), property clearance certifica	Municipality / Plot	Web Scraping / API inspection	Hidden JSON	Required (Sanad App)	Daily	None	High security / Sanad integration	Manages the *Al-Taqseem* process where 33% of land is designated for public use. Cleara

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
			tes								nces indicate transaction intent.
OpenStreetMap (OSM) Jordan	https://download.geofabrik.de/asia/jordan.html	National	Physical geography, street networks, building footprints, unverified land use	Street / Building	Bulk GeoPackage / Shapefile	No	None	Daily	https://wiki.openstreetmap.org/	None for bulk downloads	Shows physical existence rather than regulatory zoning. Highly variable data quality outside major urban centers.

Macroeconomic, Demographic, and Supply Indicators

Real estate pricing cannot be modeled in a vacuum; it is deeply intertwined with demographic shifts, inflation, and the forward-looking supply pipeline. To accurately model demand, the platform must continuously ingest macroeconomic time-series data from state statistical bodies.

Department of Statistics (DoS)

The DoS serves as the primary repository for demographic and economic metrics. Most critically for real estate, the DoS publishes monthly building permit statistics, detailing the number of licenses issued and the total licensed area in square meters (m^{2}), broken down by governorate, occupancy type, and construction material. This dataset functions as the only official forward-looking supply indicator in the country. However, platform algorithms must apply a discount factor to this data, as building permits measure licensed area, not actual construction starts; many permits in Jordan are issued but face construction delays or cancellations.
The DoS also maintains the Jordan Data Portal (jdp.dos.gov.jo), which tracks Sustainable Development Goals (SDGs), the Consumer Price Index (CPI)—including a national shelter component that tracks rental price movements—and comprehensive input-output tables for the national economy. While the most recent comprehensive census data is from 2015, the DoS

provides ongoing household statistics and poverty rate estimations.

Central Bank of Jordan (CBJ)

The Central Bank of Jordan provides the longest continuous real estate price series in the country via its Real Estate Price Index, produced jointly with the DLS. Operating on a 2012 base year, this quarterly index tracks price changes across four categories: apartments, houses, villas, and residential land.
For an advanced intelligence platform, the CBJ Index is indispensable for implementing the *Repeat Sales Index* methodology. Because the index measures price direction rather than absolute square meter values, it is used to calculate time-adjustment factors for stale comparable sales. For example, if the platform identifies a comparable transaction from 18 months ago, the CBJ index provides the exact quantitative coefficient required to adjust that historical price to present-day market realities, stripping out compositional ambiguity.

Housing and Urban Development Corporation (HUDC)

The HUDC operates as a facilitator for the organized private sector, tracking the ratio of new versus existing housing units registered in the Kingdom. HUDC datasets, often published in conjunction with the World Bank, provide deep insights into housing affordability, urbanization trends, and direct/indirect impacts of transport mobility on job access.

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
Department of Statistics (DoS) Data Portal	http://jdp.dos.gov.jo	National	Building permits (licensed area in sqm), population census, CPI (shelter component), construction activity	Governorate	Manual CSV / PDF download	No	None	Monthly / Annual	None	None	Permits indicate licensed area, not actual construction starts. Formal requests required for microdata.
Central Bank of Jordan (CBJ) Statisti	http://statisticaldb.cbj.gov.jo/index?l	National	Real Estate Price Index (Apart	Governorate	Manual Excel / CSV export	No	None	Quarterly / Monthly	None	None	Tracks price direction and magnit

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
cal Database	ang=en		ments, Houses, Villas, Land), banking credit exposure to construction								ude (Base 2012). Essential for time-adjustment of stale comparable sales.
Housing and Urban Development Corporation (HUDC)	https://www.hudc.gov.jo/Default/En	National	Housing sector reviews, urban planning reports, subsidized housing projects, auction sales	National / Regional	Manual PDF parsing / Web Scraping	No	None	Ad-hoc	None	Standard web limits	Critical for understanding government interventions in low-to-middle income housing supply.
DoS Open Data (via CKAN)	https://openda ta.gov.jo	National	Job categories by nationality, employment statistics, household demographics	Governorate	CKAN API	Yes	None	Annual	https://openda ta.gov.jo/en/api-guide/	Standard CKAN API rate limits	Demographic shifts directly impact rental demand and housing affordability modeling.

Housing Finance, Mortgages, and Capital Markets

Real estate development is a highly capital-intensive industry. The availability of liquidity, the cost of debt, and institutional investment flows dictate both the supply side (developer feasibility) and the demand side (buyer purchasing power).

Jordan Mortgage Refinance Company (JMRC)

The JMRC was established to develop the housing finance market by providing medium and long-term financing to banks. By continuously monitoring the JMRC's corporate bond issuances (e.g., Issue No. 354 yielding 5.80% for two years) and refinancing loan volumes, an intelligence platform can establish the risk-free rate and mortgage risk premiums specific to the Jordanian market.

Within the context of an Automated Valuation Model (AVM), JMRC interest rates are a critical input for the *Income Approach*. By converting borrowing costs into Capitalization Rates (Cap Rates), the platform can calculate the maximum acceptable sales price for investment-grade properties based on expected rental yields. Furthermore, JMRC's published return on investment (ROI) metrics and financial statements provide a barometer for the overall health of the mortgage sector.

Amman Stock Exchange (ASE) and Securities Depository Center (SDC)

The public equity markets provide transparency into the capitalization of Jordan's largest real estate developers and construction firms. The ASE and SDC publish exhaustive datasets to the CKAN Open Data portal regarding the ownership classification of real estate equities, traded bond volumes, and investor sector demographics.

Tracking the market capitalization and daily trading volumes of listed real estate entities (such as those analyzed in corporate governance studies) provides a macro-level indicator of institutional confidence in the Jordanian property sector. An intelligence platform can utilize this data to identify systemic capital constraints that may forecast a slowdown in project deliveries.

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
Jordan Mortgage Refinance Company (JMRC)	https://jmrc.com.jo/en/Annual-financial-statements	National	Bond issue rates, refinance loan volumes, financial statements, mortga	Institutional	PDF parsing / Web Scraping	No	None	Ad-hoc / Annual	None	Standard web limits	Dictates capital liquidity in the housing market. Bond interest rates serve

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
			ge liquidity metrics								as proxies for mortgage risk premiums.
Securities Depository Center (SDC) via CKAN	https://opendata.gov.jo/en/api/3/action/datastore_search?resource_id=[id]	National	Ownership classification of real estate equities, traded bond volumes, investor sector data	Corporate / Sector	CKAN API	Yes	None	Monthly	https://opendata.gov.jo/en/api-guide/	Standard CKAN API rate limits	Reveals institutional capital flows into listed real estate and construction development firms.
Amman Stock Exchange (ASE) Market Data	https://opendata.gov.jo/en/organization/ase-migrated_2036	National	Listed real estate developer share prices, trading volumes, corporate governance indices	Corporate	CKAN API / CSV download	Yes	None	Daily / Monthly	https://opendata.gov.jo/en/api-guide/	Standard CKAN API rate limits	Tracks the financial health and market capitalization of Jordan's largest real estate developers.
Orphans Fund Development	https://opendata.gov.jo/en/or	National	Real estate investments,	Asset Level	CKAN API / CSV	Yes	None	Annual	https://opendata.gov.jo/en/ap	Standard CKAN API	Provides insight into

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
Corporation (via CKAN)	ganization/ofdc gov-migrated_2108		buildings invested as sale or rent						i-guide/	rate limits	institutional real estate holdings and state-managed property portfolios.

Environmental Overlays, Utilities, and Climate Risk

In a highly water-scarce geography like Jordan, real estate viability is fundamentally constrained by environmental factors and utility connectivity. Properties without direct connections to municipal water or sewer networks rely on trucked water, trading at significant discounts. Consequently, utility mapping and climate resilience are non-negotiable data layers for a comprehensive intelligence platform.

Ministry of Water and Irrigation (MWI) and Water Utilities

The MWI manages an extensive GIS network mapping water utility catchments, wastewater treatment plants, groundwater recharge zones, and well locations. The MWI is currently executing a major strategic initiative to migrate its geospatial infrastructure to the ArcGIS Pro Utility Network, which will enable advanced spatial analytics and topological validation.
This federal infrastructure is supported by corporatized regional utilities, including Miyahuna (central Jordan), Yarmouk Water Company (northern governorates), and Aqaba Water (southern governorates). Miyahuna publishes open datasets detailing the exact procedures, documentation, and costs required to connect new properties to the water and sewage networks. For developers, this data is essential for calculating the true *Cost Approach* of a new build, as off-grid properties incur massive developmental penalties. Furthermore, the MWI assesses groundwater vulnerability to climate change, combining satellite imagery and DoS statistical data to map water poverty indices and adaptive capacity across the Kingdom.

Meteorological and Geological Resilience

Climate change-induced increases in temperature and flash flooding pose direct risks to real estate infrastructure in Jordan's wadis and topographical depressions. The Jordan Meteorological Department (JMD) provides historical climate data, precipitation maps, and extreme weather projections. When synthesized with elevation data, this information allows the platform to automatically flag properties located in high-risk flood zones.
Sub-surface stability is mapped by the Ministry of Energy and Mineral Resources (MEMR),

which maintains a dedicated GIS portal (memrgis.memr.gov.jo). The MEMR exposes ArcGIS REST services detailing the geology of Jordan, mineral resource locations, and oil shale investment opportunities. For large-scale land acquisitions—particularly industrial estates or residential masterplans—geological data dictates excavation and foundation construction costs.

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
Ministry of Water and Irrigation (MWI) GIS	https://gisportal.mwi.gov.jo/arcgis/rest/services	National	Water utility networks, wastewater treatment catchments, groundwater recharge zones, well locations	Infrastructure level	ArcGIS REST	Yes	Token required for secured layers	Continuous	https://gisportal.mwi.gov.jo/arcgis/rest/info	API throttling	Vital for confirming property connectivity to state utilities. Currently migrating to ArcGIS Pro Utility Network.
Miyahuna (Jordan Water Company) Open Data	https://opendata.gov.jo/en/organization/miyahuna-migrated_2153	Central Jordan	Water and sewage network connection procedures, utility statistics	Governorate	CKAN API / PDF	Yes	None	Ad-hoc	https://opendata.gov.jo/en/api-guide/	Standard CKAN API rate limits	Provides the administrative requirements and costs associated with connecting new developments to the grid.
National Electric Power	htt[span_84](start_span)[spa	National	High-voltage transmission	Regional / Infrastructure	Web Scraping / PDF	No	None	Annual	None	Standard web limits	High-voltage lines impact

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
Company (NEPCO)	n_84](end_span)ps://nepco.com.jo/en/JordanElecticSystem.aspx		lines (132kV/400kV), substation capacities, electricity tariffs, grid maps								residential property values negatively (easements) while dictating industrial feasibility.
Jordan Meteorological Department (JMD)	https://jmd.gov.jo/en	National	Historical climate data, temperature extremes, precipitation maps, weather station observations	50km x 50km Grid	Manual download / Web Scraping	No	None	Daily	None	None	Used to model climate change resilience, flood risks in wadis, and energy efficiency demands for real estate.
Ministry of Energy and Mineral Resources (MEMR) GIS	https://memrgis.memr.gov.jo/arcgis/rest/services	National	Mineral resources, petroleum investment opportunities, renewa	Zone / Plot level	ArcGIS REST	Yes	None	Periodic	https://memrgis.memr.gov.jo/arcgis/rest/info	Unknown	Critical for large-scale land acquisition strategies, particul

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
			ble energy zones								arly for industrial or solar farm developments.

Commercial Portals, Asking Prices, and Market Scraping

Because official Jordanian sources strictly anonymize unit-level transaction prices, the intelligence platform must reverse-engineer the market by aggregating *asking prices* at a massive scale. By scraping commercial real estate portals, the platform can build the foundational dataset required for a *Hedonic Pricing Model*—a statistical methodology that decomposes observed prices into the implicit values of underlying attributes (e.g., size, floor level, view, parking).

The Portal Landscape: Bayut, OpenSooq, Homes Jordan, and EurekaJO

The Jordanian online real estate landscape is dominated by a few key players. Bayut Jordan serves as the premier portal for structured, agent-listed properties, providing highly granular data points (bedrooms, building age, location, and listing history). OpenSooq operates as a high-volume classifieds platform, capturing a larger share of private sellers and mid-to-lower tier properties. Homes Jordan focuses on the curated, premium residential markets in West Amman (Abdoun, Dabouq), while EurekaJO is an emerging mobile-first platform offering virtual reality views and direct buyer-seller interactions.

Extraction Methodology and Technical Architecture

Scraping these portals is non-trivial. Modern platforms like Bayut heavily utilize JavaScript rendering and infinite scroll pagination, rendering standard HTTP requests ineffective. To successfully extract this data, the intelligence platform must deploy a headless browser architecture (utilizing tools such as Python with Playwright or Selenium). Furthermore, aggressive anti-bot protections (such as Cloudflare and PerimeterX) mandate the use of rotating residential proxies and algorithmically randomized crawl delays.

Once extracted, this data must be subjected to an absorption rate analysis. By tracking listing IDs over time, the platform can identify when properties are removed—serving as a proxy for a sale—and calculate the "Months of Supply" in a specific neighborhood. A critical caveat remains: asking prices in Jordan carry a systemic premium of 5% to 15% above actual achieved

transaction prices.

Social and Human Intelligence Networks

To calibrate the asking-price spread, the platform must integrate unstructured social data and human intelligence. Facebook public groups (searchable via terms like عقارات عمان) are highly active ecosystems where developers launch projects and private sellers list unrepresented inventory. Extracting this data requires the Facebook Graph API or third-party scrapers (e.g., Apify), followed by a Natural Language Processing (NLP) pipeline tailored to the Jordanian Arabic dialect to parse unstructured text into structured price and area fields.
Ultimately, the absolute closest proxy to the DLS's hidden transaction data resides in closed, end-to-end encrypted WhatsApp broker networks. Because these networks cannot be programmatically scraped, the intelligence platform must operationalize a structured human-intelligence workflow—employing analysts to actively monitor broker broadcasts and log achieved transaction prices manually.

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
Bayut Jordan	https://www.bayut.jo/en/	National (High Amman density)	Active asking prices, size (m^{2}), bedrooms, floor level, building age, agent info, listing history	Building / Neighborhood	Playwright / Selenium	No	None	Daily	None	High anti-bot (Cloudflare/PerimeterX), requires rotating proxies	Highest data structure quality. Agent-heavy inventory. Requires headless browser for JS rendering.
OpenSooq Real Estate	https://jo.opensooq.com/en/real-estate	National	Classifieds asking prices, sizes, seller type (individual vs agent), unstruc	Neighborhood	Playwright / Selenium	No	None	Daily	None	Pagination traps (infinite scroll), IP blocking	High volume of private sellers. Lower data quality requiring NLP/cl

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
			tured text								eaning. Complements Bayut's agent focus.
Homes Jordan	https://www.homes-jordan.com/en/	Amman (Premium)	Curated upper-market residential asking prices, verified agent contacts, monthly market blogs	Neighborhood	Web Scraping / Manual	No	None	Daily	None	Standard anti-bot	Focused on high-end Amman (Abdoun, Dabouq). Smaller inventory makes it viable for manual/light automated tracking.
Eureka JO	https://eurekajo.com	National	Listings for apartments, villas, lands, farms; VR/photo views	Neighborhood	Web Scraping	No	None	Daily	None	Standard anti-bot	Emerging platform with a mobile-first approach. Connects buyers/sellers directly, capturing

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
											unique inventory.
Facebook Public Real Estate Groups	https://www.facebook.com/search/groups/?q=عقارات عمان	National	Informal asking prices, developer launches, payment plans, market sentiment, time-to-sell	Neighborhood / Unstructured	Graph API / Apify / PhantomBuster	Yes (Graph API)	Required (Facebook account)	Live	https://developers.facebook.com/docs/graph-api	Strict API rate limits, dynamic DOM changes	Highly unstructured Arabic text. Requires custom NLP pipelines to extract price and area. captures lower-tier and private seller markets.
Broker WhatsApp Networks	N/A (Closed Networks)	Highly Localized	Real-time deal flow, achieved transaction prices, informal price reductions, unlisted inventory	Deal level	Manual request (Human Intelligence)	No	Invitation Only	Live	None	N/A (End-to-End Encrypted)	The *only* source of true transaction prices in Jordan. Requires relationship building and structured

Source Name	URL	Coverage (Jordan)	Data Available	Granularity	Extraction Method	API Available?	Authentication	Update Frequency	Documentation URL	Rate Limits / Anti-Bot	Notes
											manual logging.

Analytical Synthesis and AVM Deployment Strategy

The exhaustive compilation of these data sources is merely the foundation; the actualization of a real estate intelligence platform requires the algorithmic harmonization of these disparate vectors into a cohesive Automated Valuation Model (AVM).

An AVM for the Jordanian market must operate as a system of interconnected mathematical models, circumventing the lack of official transaction price transparency. The architecture relies on three foundational pillars:

1. **The Hedonic Pricing Engine:** By scraping the vast inventory from Bayut, OpenSooq, and EurekaJO, the platform utilizes regression mathematics to isolate the implicit value of every physical attribute of a property. In this model, the price is not treated as a single integer, but as a bundle of coefficients: Price = \beta_0 + \beta_1(Size) + \beta_2(Floor) + \beta_3(Location) + \epsilon.
2. **The Spatial Intersection Matrix:** To solve for the "Location" coefficient (\beta_3), the platform must continuously ping the ArcGIS REST endpoints of the GAM, ASEZA, and MPWH. By executing programmatic spatial joins against the geocodes extracted from property listings, the AVM algorithmically enriches the hedonic model with regulatory parameters (e.g., FAR, proximity to the MWI water network, positioning outside of JMD-identified flood zones). This is the critical step where machine learning models—specifically gradient boosting frameworks like XGBoost or LightGBM—excel, identifying the non-linear interaction effects between a high floor and a specific Amman municipal zone.
3. **Temporal Calibration and Residual Logic:** Because the Hedonic engine is trained on asking prices, it inherently overvalues the market. The final algorithmic layer must apply a discount matrix calibrated by the human-intelligence data extracted from the WhatsApp broker networks. Concurrently, the CBJ Real Estate Price Index acts as a temporal calibrator, mathematically adjusting historical comparables to current market conditions via repeat-sales indexing logic. Finally, the MoLA subdivision rules (*Al-Taqseem*) and DoS building permit pipelines feed into a *Cost Approach* module, calculating the maximum residual land value a developer can rationally pay for a vacant plot given current material costs and forward-looking supply saturation.

By systematically maintaining these extraction pipelines—adapting to the dynamic DOM changes of commercial portals and adhering to the authentication protocols of the Jordanian CKAN and ArcGIS web services—the intelligence platform will transcend traditional intuitive pricing, establishing a scientific, data-driven hegemony over the Jordanian real estate sector.

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